

# Radial Extremality for LRU Caching and the Fill–Holst Conjecture

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## Abstract

For the independent reference model with popularity vector  $p \in \Delta_N^\circ$ , let  $H_C(p)$  denote the exact stationary hit rate of an LRU cache of capacity  $C$ . We prove that, for every  $1 \leq C < N$ , the uniform popularity vector is the unique global minimizer of  $H_C$  on the interior simplex. More sharply, along every nonconstant segment from the uniform vector to an interior point, the LRU hit rate is strictly increasing. The proof uses the standard exponential-age representation of the stationary LRU cache and gives an explicit positive pair-square formula for the radial derivative. Equivalently, for the move-to-front rule, the stationary search-cost distribution improves strictly in the usual stochastic order along every nonconstant ray away from uniform. This proves the radial restriction of the Fill–Holst Schur-concavity conjecture for move-to-front search-cost tails. In particular, all LRU miss probabilities and all nonconstant nondecreasing stack-depth costs decrease strictly along such rays. The result is radial rather than Schur-convex: full majorization monotonicity for LRU is known to fail, and the proof identifies the special positivity that survives on rays from the uniform vector.

## 1 Introduction

The independent reference model (IRM) is the classical model in which successive requests are independent and item  $i$  is requested with probability  $p_i$  in discrete time, equivalently with rate  $p_i$  in the rate-one Poissonized model. Under LRU, the stationary stack order has an exact exponential-age representation: if  $A_1, \dots, A_N$  are independent exponential random variables with  $A_i \sim \text{Exp}(p_i)$ , then the stationary cache of capacity  $C$  consists of the  $C$  items with smallest ages. This representation follows by Poissonizing the request sequence into independent Poisson streams of rates  $p_i$  and using their backward recurrence times; conditioning on the next arrival recovers the original discrete request distribution. The hit rate is therefore an exact finite-dimensional function of the popularity vector, not a characteristic-time or Che-type approximation. Exact formulae and asymptotic approximations for LRU under the IRM go back to King, Fagin, Fagin–Price, and Flajolet–Gardy–Thimonier [9, 3, 4, 6]; Che–Tung–Wang introduced a now-standard characteristic-time approximation for hierarchical Web caching [2], Jelenković studied asymptotic MTF/LRU fault probabilities for broad popularity classes [8], and Berthet discusses the identity of the King and Flajolet–Gardy–Thimonier formulae for exact LRU miss-rate computation [1].

A natural monotonicity question asks whether making the popularity vector more uneven must improve the LRU hit rate. In the language of majorization [11], this would assert Schur-convexity of the hit rate, or equivalently Schur-concavity of the miss rate. A closely related move-to-front literature considered the same distributional question in search-cost language. Under independent requests, the stationary move-to-front list is the LRU recency order, and the LRU miss probability for a cache of capacity  $C$  is the tail probability that the move-to-front search cost exceeds  $C$ . Fill and Holst studied the stationary move-to-front search-cost distribution and its cache-fault interpretation,

and Hildebrand later addressed their conjecture that each search-cost tail is Schur-concave in the popularity vector [5, 7]. Hildebrand disproved that conjecture.

Thus full Schur-concavity is too strong even for the natural MTF/LRU tail probabilities. Vanichpun and Makowski also showed, in a broader caching setting, that majorization monotonicity may fail under LRU [12, 10]. The theorem below identifies a sharp positive remnant of that false global principle: although arbitrary majorization directions can move the LRU hit rate in the wrong direction, every straight-line move away from the uniform popularity vector strictly increases the hit rate. Equivalently, the radial restriction of the Fill–Holst tail-concavity conjecture is true for MTF/LRU.

The proof is deliberately finite and exact. We first derive a residual subset expansion for  $H_C(p)$ . A variance identity on each residual subset converts that expansion into a pair-square decomposition around the uniform vector. Radial differentiation then produces an explicit pair-square formula

$$\frac{d}{d\theta} H_C(u + \theta(q - u)) = \frac{1}{N\theta} \sum_{1 \leq a < b \leq N} (p_a(\theta) - p_b(\theta))^2 K_{ab}^{(C)}(p(\theta)),$$

where every kernel  $K_{ab}^{(C)}$  is strictly positive. The positivity of the kernels is proved by converting the alternating subset sums into positive product integrals. This gives a local certificate for radial monotonicity, not merely a global comparison.

After proving the main theorem, we record its distributional consequences for the full LRU stack-depth distribution, or equivalently for the stationary move-to-front search-cost distribution. We also include an appendix with an independent occupancy-Jacobian proof. That proof explains at the differential level why radial monotonicity is compatible with failure of Schur-convexity: the asymmetric occupancy Jacobian is not positive in every tangent direction, but its sign-indefinite minor collapses to a positive square along rays from the uniform vector.

## 2 Setup and main results

Let

$$\Delta_N^\circ = \left\{ p = (p_1, \dots, p_N) : p_i > 0, \sum_{i=1}^N p_i = 1 \right\}, \quad u = \left( \frac{1}{N}, \dots, \frac{1}{N} \right).$$

For  $R \subseteq [N] := \{1, \dots, N\}$ , write

$$p_R = \sum_{i \in R} p_i, \quad p_\emptyset = 0.$$

Let  $A_1, \dots, A_N$  be independent exponential random variables with  $A_i \sim \text{Exp}(p_i)$ , and let  $S_C$  be the set of the  $C$  indices with smallest ages. The exact stationary LRU hit rate is

$$H_C(p) = \mathbb{E} \left[ \sum_{i \in S_C} p_i \right]. \tag{1}$$

Equivalently, since the next request has distribution  $p$  independently of the stationary stack,  $H_C(p)$  is the stationary probability that the next requested item is already in an LRU cache of capacity  $C$ .

**Theorem 2.1** (Radial LRU extremality). *Let  $N \geq 2$ , let  $1 \leq C < N$ , and let  $q \in \Delta_N^\circ$  be nonuniform. Define*

$$p(\theta) = u + \theta(q - u), \quad 0 \leq \theta \leq 1.$$

Then

$$\frac{d}{d\theta} H_C(p(\theta)) > 0, \quad 0 < \theta \leq 1.$$

Consequently  $\theta \mapsto H_C(p(\theta))$  is strictly increasing on  $(0, 1]$ .

**Corollary 2.2** (Uniform popularity is the unique worst case). *Let  $N \geq 2$  and  $1 \leq C < N$ . For every nonuniform  $p \in \Delta_N^\circ$ ,*

$$H_C(p) > H_C(u) = \frac{C}{N}.$$

*Thus  $u$  is the unique global minimizer of the LRU hit rate on  $\Delta_N^\circ$ . Equivalently, the LRU miss rate  $1 - H_C(p)$  is uniquely maximized at  $u$  on  $\Delta_N^\circ$ .*

*Proof.* Apply Theorem 2.1 with  $q = p$ . Then  $p(0) = u$  and  $p(1) = p$ . By Lemma 3.1,  $H_C$  is continuous along this segment, and Theorem 2.1 gives strict monotonicity on  $(0, 1]$ ; hence  $H_C(p) > H_C(u)$ . Under  $u$ , symmetry gives  $\mathbb{P}(i \in S_C) = C/N$  for every item  $i$ , and therefore

$$H_C(u) = \sum_{i=1}^N \frac{1}{N} \cdot \frac{C}{N} = \frac{C}{N}.$$

□

The proof of Theorem 2.1 proceeds through a stronger positive kernel formula. Put

$$L = N - C + 1.$$

Thus  $2 \leq L \leq N$ . For  $a < b$  and  $p \in \Delta_N^\circ$ , define

$$K_{ab}^{(C)}(p) = N \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} (-1)^{|R|-L} \binom{|R|-2}{|R|-L} \left( \frac{1}{|R| p_R} + \frac{1}{N p_R^2} \right). \quad (2)$$

**Theorem 2.3** (Positive pair-kernel derivative formula). *For every  $p \in \Delta_N^\circ$  and every  $a < b$ ,*

$$K_{ab}^{(C)}(p) > 0.$$

*Moreover, if  $p(\theta) = u + \theta(q - u)$  with  $0 < \theta \leq 1$ , then*

$$\frac{d}{d\theta} H_C(p(\theta)) = \frac{1}{N\theta} \sum_{1 \leq a < b \leq N} (p_a(\theta) - p_b(\theta))^2 K_{ab}^{(C)}(p(\theta)). \quad (3)$$

Theorem 2.1 follows immediately from Theorem 2.3: if  $q \neq u$  and  $0 < \theta \leq 1$ , then  $p(\theta)$  is nonuniform, so at least one pair difference in (3) is nonzero.

### 3 A residual-subset formula for the hit rate

We first convert the exponential-clock representation into a finite alternating subset expansion. This is the only place where the cache capacity enters in an essential way.

**Lemma 3.1** (Residual-subset expansion). *Let  $1 \leq C < N$  and  $L = N - C + 1$ . For  $L \leq m \leq N$ , set*

$$\alpha_m = (-1)^{m-L} \binom{m-2}{m-L}.$$

*Then, for every  $p \in \Delta_N^\circ$ ,*

$$H_C(p) = \sum_{m=L}^N \alpha_m \sum_{\substack{R \subseteq [N] \\ |R|=m}} \frac{\sum_{i \in R} p_i^2}{p_R}. \quad (4)$$

*Proof.* Fix an item  $i$ . Its contribution to the hit rate is  $p_i \mathbb{P}(i \in S_C)$ . Conditioning on  $A_i = t$  gives

$$p_i \mathbb{P}(i \in S_C) = p_i^2 \int_0^\infty e^{-p_i t} \sum_{\substack{S \subseteq [N] \setminus \{i\} \\ |S| \leq C-1}} \prod_{j \in S} (1 - e^{-p_j t}) \prod_{\ell \notin S \cup \{i\}} e^{-p_\ell t} dt. \quad (5)$$

Here  $S$  is the set of other items whose ages are smaller than  $t$ ; item  $i$  is in the cache precisely when  $|S| \leq C - 1$ .

Expand

$$\prod_{j \in S} (1 - e^{-p_j t}) = \sum_{A \subseteq S} (-1)^{|A|} e^{-p_A t}.$$

For a term indexed by  $A \subseteq S$ , put  $B = S \setminus A$ . The total exponent in (5) is

$$p_i + p_A + p_{[N] \setminus (S \cup \{i\})} = 1 - p_B.$$

Now fix  $B \subseteq [N] \setminus \{i\}$  with  $b = |B| \leq C - 1$ . The set  $A$  may be any subset of  $[N] \setminus (\{i\} \cup B)$  with  $|A| \leq C - 1 - b$ . Hence the coefficient of  $e^{-(1-p_B)t}$  is

$$\sum_{h=0}^{C-1-b} (-1)^h \binom{N-1-b}{h}.$$

Since  $C < N$ , the elementary identity

$$\sum_{h=0}^q (-1)^h \binom{M}{h} = (-1)^q \binom{M-1}{q}, \quad 0 \leq q < M,$$

shows that this coefficient is

$$(-1)^{C-1-b} \binom{N-2-b}{C-1-b}.$$

Therefore

$$p_i \mathbb{P}(i \in S_C) = p_i^2 \sum_{\substack{B \subseteq [N] \setminus \{i\} \\ |B| \leq C-1}} (-1)^{C-1-|B|} \binom{N-2-|B|}{C-1-|B|} \frac{1}{1-p_B}.$$

Set  $R = [N] \setminus B$ . Then  $i \in R$ ,  $p_R = 1 - p_B$ , and  $m := |R| = N - |B|$  satisfies  $m \geq N - C + 1 = L$ . Moreover

$$(-1)^{C-1-|B|} \binom{N-2-|B|}{C-1-|B|} = (-1)^{m-L} \binom{m-2}{m-L} = \alpha_m.$$

Summing over  $i$  and grouping by  $R$  gives (4).  $\square$

## 4 Pair-square decomposition

The residual expansion separates cleanly into the uniform contribution and pairwise fluctuations.

**Lemma 4.1** (Pair-square expansion of  $H_C$ ). *For every  $p \in \Delta_N^\circ$ ,*

$$H_C(p) = \frac{C}{N} + \sum_{1 \leq a < b \leq N} (p_a - p_b)^2 J_{ab}^{(C)}(p), \quad (6)$$

where

$$J_{ab}^{(C)}(p) = \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} (-1)^{|R|-L} \binom{|R|-2}{|R|-L} \frac{1}{|R| p_R}. \quad (7)$$

*Proof.* For any nonempty finite set  $R$  with  $m = |R|$ ,

$$\sum_{i \in R} p_i^2 = \frac{p_R^2}{m} + \frac{1}{m} \sum_{\substack{a < b \\ a, b \in R}} (p_a - p_b)^2. \quad (8)$$

This follows from

$$\sum_{\substack{a < b \\ a, b \in R}} (p_a - p_b)^2 = m \sum_{i \in R} p_i^2 - p_R^2.$$

Substituting (8) into (4) gives

$$\begin{aligned} H_C(p) &= \sum_{m=L}^N \alpha_m \sum_{\substack{R \subseteq [N] \\ |R|=m}} \frac{p_R}{m} \\ &\quad + \sum_{1 \leq a < b \leq N} (p_a - p_b)^2 \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} \frac{\alpha_{|R|}}{|R| p_R}. \end{aligned}$$

The second term is the pair-square term in (6). The first term is constant on the simplex, because

$$\sum_{\substack{R \subseteq [N] \\ |R|=m}} p_R = \binom{N-1}{m-1} \sum_{i=1}^N p_i = \binom{N-1}{m-1}.$$

To identify the constant, evaluate at  $p = u$ . By symmetry,  $H_C(u) = C/N$ , and at  $p = u$  all pair differences vanish. Hence the constant is  $C/N$ .  $\square$

## 5 Radial differentiation

Let  $q \in \Delta_N^\circ$  and set

$$p(\theta) = u + \theta(q - u), \quad 0 < \theta \leq 1.$$

For readability, write  $x = p(\theta)$  during the differentiation. If  $R \subseteq [N]$  and  $m = |R|$ , then

$$\theta \frac{dx_i}{d\theta} = x_i - \frac{1}{N}, \quad \theta \frac{dx_R}{d\theta} = x_R - \frac{m}{N}, \quad \theta \frac{d}{d\theta}(x_a - x_b) = x_a - x_b. \quad (9)$$

Consequently,

$$\begin{aligned}\theta \frac{d}{d\theta} \left( \frac{(x_a - x_b)^2}{mx_R} \right) &= (x_a - x_b)^2 \left( \frac{2}{mx_R} - \frac{x_R - m/N}{mx_R^2} \right) \\ &= (x_a - x_b)^2 \left( \frac{1}{mx_R} + \frac{1}{Nx_R^2} \right).\end{aligned}\tag{10}$$

Differentiating (6) term by term and using (10) gives

$$\frac{d}{d\theta} H_C(p(\theta)) = \frac{1}{\theta} \sum_{1 \leq a < b \leq N} (x_a - x_b)^2 \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} (-1)^{|R|-L} \binom{|R|-2}{|R|-L} \left( \frac{1}{|R|x_R} + \frac{1}{Nx_R^2} \right).$$

This is precisely (3), with  $K_{ab}^{(C)}$  defined by (2). It remains only to prove that these kernels are strictly positive.

## 6 Strict positivity of the pair kernels

Fix  $p \in \Delta_N^\circ$  and a pair  $a < b$ . Let

$$T = [N] \setminus \{a, b\}, \quad s = p_a + p_b, \quad r = L - 2 = N - C - 1.$$

Since  $1 \leq C < N$ , we have  $0 \leq r \leq N - 2 = |T|$ . Writing  $R = \{a, b\} \cup U$  with  $U \subseteq T$ , the coefficient in (2) becomes

$$(-1)^{|R|-L} \binom{|R|-2}{|R|-L} = (-1)^{|U|-r} \binom{|U|}{r},$$

where the binomial coefficient is zero if  $|U| < r$ . Hence

$$\frac{1}{N} K_{ab}^{(C)}(p) = \Phi_r + \frac{1}{N} \Psi_r,\tag{11}$$

where

$$\Phi_r := \sum_{U \subseteq T} (-1)^{|U|-r} \binom{|U|}{r} \frac{1}{(|U|+2)(s+p_U)},\tag{12}$$

$$\Psi_r := \sum_{U \subseteq T} (-1)^{|U|-r} \binom{|U|}{r} \frac{1}{(s+p_U)^2}.\tag{13}$$

We prove that both  $\Phi_r$  and  $\Psi_r$  are strictly positive.

For  $t \geq 0$  and  $0 \leq y \leq 1$ , define

$$B_r(y, t) = \sum_{U \subseteq T} (-1)^{|U|-r} \binom{|U|}{r} y^{|U|} e^{-p_U t}.\tag{14}$$

Let

$$P_t(y) = \prod_{j \in T} (1 - ye^{-p_j t}).$$

Since

$$P_t(y) = \sum_{U \subseteq T} (-1)^{|U|} y^{|U|} e^{-p_U t},$$

we have

$$B_r(y, t) = \frac{(-1)^r y^r}{r!} P_t^{(r)}(y). \quad (15)$$

Differentiating the product gives the positive form

$$B_r(y, t) = y^r \sum_{\substack{W \subseteq T \\ |W|=r}} e^{-p_W t} \prod_{j \in T \setminus W} (1 - ye^{-p_j t}). \quad (16)$$

For  $t > 0$  and  $0 < y \leq 1$ , every factor  $1 - ye^{-p_j t}$  is strictly positive, because  $p_j > 0$ . Therefore

$$B_r(y, t) > 0, \quad t > 0, \quad 0 < y \leq 1. \quad (17)$$

Now use the elementary integral identities

$$\frac{1}{s + p_U} = \int_0^\infty e^{-(s+p_U)t} dt, \quad \frac{1}{|U| + 2} = \int_0^1 y^{|U|+1} dy,$$

and

$$\frac{1}{(s + p_U)^2} = \int_0^\infty t e^{-(s+p_U)t} dt.$$

Since all sums are finite, we may interchange sums and integrals. From (12) and (14),

$$\Phi_r = \int_0^\infty e^{-st} \int_0^1 y B_r(y, t) dy dt.$$

By (17), the integrand is strictly positive for  $t > 0$  and  $0 < y < 1$ . Hence  $\Phi_r > 0$ . Similarly, from (13) and (14),

$$\Psi_r = \int_0^\infty t e^{-st} B_r(1, t) dt > 0.$$

Combining these inequalities with (11) gives  $K_{ab}^{(C)}(p) > 0$ . This proves Theorem 2.3, and hence Theorem 2.1.

*Remark 6.1* (Endpoint capacities). For  $C = 1$ , one has  $L = N$ , and the residual formula reduces to

$$H_1(p) = \sum_{i=1}^N p_i^2.$$

For  $C = N$ , the hit rate is identically 1, so strict radial monotonicity is not a meaningful statement. The theorem therefore covers exactly the nontrivial range  $1 \leq C < N$ .

## 7 Move-to-front search costs and miss-probability consequences

The radial theorem has a useful distributional interpretation. Under independent requests, the stationary move-to-front list is the same recency ordering used by LRU, namely the ordering of all  $N$  items from most recently requested to least recently requested. Let  $D = D(p)$  be the stationary move-to-front search cost, equivalently the LRU stack depth of the next requested item in stationarity. Then  $D \in \{1, \dots, N\}$ , and a cache of capacity  $C$  hits exactly when  $D \leq C$ . Therefore

$$\mathbb{P}_p(D \leq C) = H_C(p), \quad 1 \leq C \leq N. \quad (18)$$

Here  $H_N \equiv 1$ , while the nontrivial thresholds are  $1 \leq C < N$ .

**Corollary 7.1** (Radial stochastic improvement for move-to-front search cost). *Let  $N \geq 2$ , let  $q \in \Delta_N^\circ$  be nonuniform, and put  $p(\theta) = u + \theta(q - u)$ . If  $0 < \theta_1 < \theta_2 \leq 1$ , then the move-to-front search cost under  $p(\theta_2)$  is strictly smaller than the move-to-front search cost under  $p(\theta_1)$  in the usual stochastic order: for every  $1 \leq C < N$ ,*

$$\mathbb{P}_{p(\theta_2)}(D \leq C) > \mathbb{P}_{p(\theta_1)}(D \leq C),$$

or equivalently

$$\mathbb{P}_{p(\theta_2)}(D > C) < \mathbb{P}_{p(\theta_1)}(D > C).$$

*Proof.* Identity (18) gives  $\mathbb{P}_{p(\theta)}(D \leq C) = H_C(p(\theta))$ . For  $1 \leq C < N$ , Theorem 2.1 shows that this quantity is strictly increasing in  $\theta$ . Taking complements gives the tail form.  $\square$

**Corollary 7.2** (Nondecreasing search-cost functionals). *Under the hypotheses of Corollary 7.1, for every nondecreasing function  $g : \{1, \dots, N\} \rightarrow \mathbb{R}$ ,*

$$\mathbb{E}_{p(\theta_2)}g(D) \leq \mathbb{E}_{p(\theta_1)}g(D).$$

*The inequality is strict whenever  $g$  is not constant. In particular, the LRU miss probability*

$$M_C(p) := 1 - H_C(p) = \mathbb{P}_p(D > C)$$

*is strictly decreasing along every nonconstant ray away from uniform for each  $1 \leq C < N$ , and the expected move-to-front search cost*

$$\mathbb{E}_p D = 1 + \sum_{c=1}^{N-1} M_c(p)$$

*is strictly decreasing along every such ray.*

*Proof.* Use the discrete tail-sum decomposition

$$\mathbb{E}g(D) = g(1) + \sum_{c=1}^{N-1} (g(c+1) - g(c))\mathbb{P}(D > c). \quad (19)$$

If  $g$  is nondecreasing, every coefficient in the sum is nonnegative, and Corollary 7.1 gives the asserted monotonicity. If  $g$  is not constant, at least one coefficient  $g(c+1) - g(c)$  is positive, and the corresponding tail probability decreases strictly. The statements about miss probability and expected search depth are the special cases  $g = \mathbf{1}_{\{C+1, \dots, N\}}$  and  $g(d) = d$ .  $\square$

In this language, Theorem 2.1 proves the radial restriction of the Fill–Holst tail-concavity conjecture. Hildebrand’s counterexample shows that full Schur-concavity of the search-cost tails is false. The present result shows that the conjectured monotonicity nevertheless survives on every segment emanating from the uniform distribution.

## 8 Why the theorem is radial, not Schur-convex

The preceding result should not be interpreted as a disguised Schur-convexity theorem. Full Schur-convexity would require monotonicity under all elementary majorization transfers, not merely along rays from the uniform vector. The Fill–Holst tail conjecture is false [7], and the known Vanichpun–Makowski examples show that LRU can violate the corresponding majorization monotonicity [12, 10].

The point is that the radial direction from uniform aligns with a local asymmetry in the LRU occupancy map.

Appendix A gives the details, but the local mechanism is simple to state. Let  $\pi_k(\lambda)$  denote the probability that item  $k$  is present in the stationary LRU cache when the exponential rates are  $\lambda$ . For  $i \neq k$ , the off-diagonal sensitivity has the form

$$\frac{\partial \pi_k}{\partial \lambda_i}(\lambda) = -\lambda_k G_{ik}(\lambda), \quad G_{ik}(\lambda) = G_{ki}(\lambda) > 0.$$

Thus the occupancy Jacobian is generally asymmetric. For an arbitrary tangent direction  $\delta$  with  $\sum_i \delta_i = 0$ , the occupancy-sensitivity part of the directional derivative contains

$$\sum_{i < k} G_{ik}(\lambda)(\lambda_i - \lambda_k)(\lambda_k \delta_i - \lambda_i \delta_k),$$

and the minor  $\lambda_k \delta_i - \lambda_i \delta_k$  has no fixed sign. Along the radial path  $\lambda = u + \theta(q - u)$ , however,

$$\delta_i = \frac{\lambda_i - 1/N}{\theta}, \quad \lambda_k \delta_i - \lambda_i \delta_k = \frac{\lambda_i - \lambda_k}{N\theta},$$

so the sign-indefinite term collapses to the positive square

$$\frac{1}{N\theta} \sum_{i < k} G_{ik}(\lambda)(\lambda_i - \lambda_k)^2.$$

This is the local reason radial monotonicity survives even though full Schur-convexity fails.

## 9 Further directions

The distinction between global Schur-concavity and radial monotonicity suggests a broader structural problem. For policies such as RANDOM replacement and FIFO, stronger Schur-concavity results are already known under the IRM, so radial monotonicity follows immediately [12, 10]. LRU/MTF is different: full Schur-concavity fails, but the present paper proves strict radial monotonicity. This raises the corresponding radial question for other self-organizing policies for which full Schur-concavity fails or remains unknown. Identifying the finite-state structure that controls radial monotonicity for such policies is left for future work.

## A An occupancy-Jacobian proof

This appendix gives an independent proof of Theorem 2.1. The main proof above is shorter, but the Jacobian proof isolates the local mechanism behind the radial positivity.

Fix  $\lambda \in \mathbb{R}_{>0}^N$ , and let  $A_i \sim \text{Exp}(\lambda_i)$  be independent. Let  $S_C(\lambda)$  be the set of the  $C$  indices with smallest ages, and define

$$\pi_k(\lambda) = \mathbb{P}(k \in S_C(\lambda)).$$

The map  $\pi_k$  is homogeneous of degree zero, and

$$\sum_{k=1}^N \pi_k(\lambda) = C, \quad H_C(p) = \sum_{k=1}^N p_k \pi_k(p). \quad (20)$$

The first identity holds because  $|S|_C = C$  deterministically. Differentiating it gives

$$\sum_{k=1}^N \frac{\partial \pi_k}{\partial \lambda_i}(\lambda) = 0 \quad \text{for every } i. \quad (21)$$

**Lemma A.1** (Occupancy sensitivity). *For  $i \neq k$  and any  $\lambda \in \mathbb{R}_{>0}^N$ ,*

$$\frac{\partial \pi_k}{\partial \lambda_i}(\lambda) = -\lambda_k G_{ik}(\lambda), \quad (22)$$

where

$$G_{ik}(\lambda) = \int_0^\infty t e^{-(\lambda_i + \lambda_k)t} \mathbb{P}(M_{ik}(t) = C - 1) dt,$$

and

$$M_{ik}(t) = \#\{j \notin \{i, k\} : A_j < t\}.$$

The kernel is symmetric and nonnegative,  $G_{ik} = G_{ki} \geq 0$ . If  $1 \leq C < N$ , then  $G_{ik} > 0$  for every pair  $i \neq k$ .

*Proof.* Conditioning on  $A_k = t$  gives

$$\pi_k(\lambda) = \int_0^\infty \lambda_k e^{-\lambda_k t} \Phi_k(t; \lambda) dt,$$

where

$$\Phi_k(t; \lambda) = \mathbb{P} \left( \sum_{j \neq k} \mathbf{1}_{\{A_j < t\}} \leq C - 1 \right).$$

For  $i \neq k$ , write  $M_{ik}(t) = \sum_{j \notin \{i, k\}} \mathbf{1}_{\{A_j < t\}}$ . Splitting off the  $i$ th indicator and putting  $x_i(t) = 1 - e^{-\lambda_i t}$ ,

$$\Phi_k = \mathbb{P}(M_{ik} \leq C - 1) - x_i(t) \mathbb{P}(M_{ik} = C - 1).$$

Hence

$$\frac{\partial \Phi_k}{\partial \lambda_i} = -t e^{-\lambda_i t} \mathbb{P}(M_{ik}(t) = C - 1).$$

Differentiating under the integral is justified by dominated convergence, for example on compact subsets of  $\mathbb{R}_{>0}^N$ : the differentiated integrand is bounded by an integrable multiple of  $t e^{-\eta t}$  for some  $\eta > 0$ . This gives (22). The symmetry  $G_{ik} = G_{ki}$  is immediate from the integral expression. For strict positivity, note that when  $1 \leq C < N$ , we have  $0 \leq C - 1 \leq N - 2$ . For every  $t > 0$ , each event probability in the Poisson-binomial mass  $\mathbb{P}(M_{ik}(t) = C - 1)$  is a nonempty sum of strictly positive products, so the integrand is strictly positive on  $(0, \infty)$ .  $\square$

The diagonal entries are determined by (21):

$$\frac{\partial \pi_i}{\partial \lambda_i}(\lambda) = \sum_{k \neq i} \lambda_k G_{ik}(\lambda). \quad (23)$$

We also need the elementary monotonicity of occupancy probabilities.

**Lemma A.2** (Ordering law). *For every permutation  $\sigma = (\sigma_1, \dots, \sigma_N)$  of  $[N]$ ,*

$$\mathbb{P}(A_{\sigma_1} < A_{\sigma_2} < \dots < A_{\sigma_N}) = \prod_{r=1}^N \frac{\lambda_{\sigma_r}}{\sum_{m=r}^N \lambda_{\sigma_m}}.$$

*Equivalently, the exponential-clock order is generated by repeatedly choosing the next index from the remaining indices with probability proportional to its rate.*

*Proof.* The first clock to ring has label  $j$  with probability  $\lambda_j / \sum_m \lambda_m$ . Conditional on that first label, the residual waiting times of the remaining clocks are again independent exponentials with their original rates, by memorylessness. Iterating gives the displayed product.  $\square$

**Lemma A.3** (Occupancy monotonicity). *If  $\lambda_k \geq \lambda_i$ , then  $\pi_k(\lambda) \geq \pi_i(\lambda)$ . Consequently,*

$$(\lambda_a - \lambda_b)(\pi_a(\lambda) - \pi_b(\lambda)) \geq 0 \quad \text{for all } a, b.$$

*Proof.* It suffices to compare the cases in which exactly one of  $i, k$  belongs to the first  $C$  positions of the exponential-clock order, since the cases in which both are cached or both are not cached cancel in  $\pi_k - \pi_i$ . Let  $\mathcal{A}$  be the set of orderings in which  $i$  appears in one of the first  $C$  positions and  $k$  appears after position  $C$ , and let  $\mathcal{B}$  be the analogous set with  $i$  and  $k$  interchanged. Swapping the labels  $i$  and  $k$  gives a bijection  $\mathcal{A} \rightarrow \mathcal{B}$ .

Fix  $\sigma \in \mathcal{A}$ . Suppose  $i$  occurs at position  $r \leq C$  and  $k$  occurs at position  $s > C$ . Let  $\tau$  be obtained from  $\sigma$  by swapping the labels  $i$  and  $k$ , and put  $d = \lambda_k - \lambda_i \geq 0$ . In the product formula of Lemma A.2, the numerator factors involving  $i$  and  $k$  cancel in the ratio  $\mathbb{P}(\tau)/\mathbb{P}(\sigma)$ , and all denominator factors outside positions  $r + 1, \dots, s$  are identical. For  $m = r + 1, \dots, s$ , the remaining-rate denominator under  $\tau$  is smaller by exactly  $d$  than the corresponding denominator under  $\sigma$ . Hence

$$\frac{\mathbb{P}(\tau)}{\mathbb{P}(\sigma)} = \prod_{m=r+1}^s \frac{D_m}{D_m - d} \geq 1,$$

where  $D_m$  denotes the corresponding remaining-rate denominator under  $\sigma$ . Summing over the paired orderings gives  $\pi_k \geq \pi_i$ .  $\square$

**Corollary A.4** (Uniform lower bound). *For every  $p \in \Delta_N^\circ$ ,*

$$H_C(p) \geq \frac{C}{N}.$$

*Proof.* Using (20),

$$H_C(p) - \frac{C}{N} = \sum_k \left( p_k - \frac{1}{N} \right) \pi_k(p) = \frac{1}{N} \sum_{a < b} (p_a - p_b)(\pi_a(p) - \pi_b(p)).$$

Every summand is nonnegative by Lemma A.3.  $\square$

**Theorem A.5** (Jacobian master identity). *Let  $N \geq 2$ ,  $1 \leq C < N$ , and let  $q \in \Delta_N^\circ$  be nonuniform. Put  $\lambda(\theta) = u + \theta(q - u)$  and  $\mathcal{H}(\theta) = H_C(\lambda(\theta))$ . Then, for  $0 < \theta \leq 1$ ,*

$$\mathcal{H}'(\theta) = \frac{1}{\theta} \left( H_C(\lambda(\theta)) - \frac{C}{N} \right) + \frac{1}{N\theta} \sum_{1 \leq i < k \leq N} G_{ik}(\lambda(\theta)) (\lambda_i(\theta) - \lambda_k(\theta))^2. \quad (24)$$

*In particular  $\mathcal{H}'(\theta) > 0$  for  $0 < \theta \leq 1$ .*

*Proof.* Write  $\lambda = \lambda(\theta)$  and  $\dot{\lambda}_i = q_i - 1/N$ . By the chain rule and (20),

$$\mathcal{H}'(\theta) = \sum_k \dot{\lambda}_k \pi_k + \sum_k \lambda_k \sum_i \frac{\partial \pi_k}{\partial \lambda_i} \dot{\lambda}_i =: T_1 + T_2.$$

Since  $\dot{\lambda}_k = (\lambda_k - 1/N)/\theta$ , the first term is

$$T_1 = \frac{1}{\theta} \sum_k \left( \lambda_k - \frac{1}{N} \right) \pi_k = \frac{1}{\theta} \left( H_C(\lambda) - \frac{C}{N} \right).$$

For the second term, use Lemma A.1 and (23):

$$\begin{aligned} T_2 &= \sum_i \lambda_i \dot{\lambda}_i \sum_{k \neq i} \lambda_k G_{ik} - \sum_{i \neq k} \lambda_k^2 \dot{\lambda}_i G_{ik} \\ &= \sum_{i \neq k} G_{ik} \lambda_k \dot{\lambda}_i (\lambda_i - \lambda_k). \end{aligned}$$

Symmetrizing over  $i \leftrightarrow k$  gives

$$T_2 = \sum_{i < k} G_{ik} (\lambda_i - \lambda_k) (\lambda_k \dot{\lambda}_i - \lambda_i \dot{\lambda}_k).$$

Along the radial path,

$$\lambda_k \dot{\lambda}_i - \lambda_i \dot{\lambda}_k = \frac{1}{\theta} \left[ \lambda_k \left( \lambda_i - \frac{1}{N} \right) - \lambda_i \left( \lambda_k - \frac{1}{N} \right) \right] = \frac{\lambda_i - \lambda_k}{N\theta}.$$

Thus

$$T_2 = \frac{1}{N\theta} \sum_{i < k} G_{ik} (\lambda_i - \lambda_k)^2.$$

Adding  $T_1$  gives (24). The first term is nonnegative by Corollary A.4. The second term is strictly positive because  $G_{ik} > 0$  for all pairs and  $\lambda(\theta)$  is nonuniform when  $0 < \theta \leq 1$ . Hence  $\mathcal{H}'(\theta) > 0$ .  $\square$

*Remark A.6* (Relation to the main kernel). The appendix proof yields a different positive decomposition of the same radial derivative. It splits the derivative into a nonnegative uniform lower-bound term and a strictly positive occupancy-sensitivity square. The main proof instead packages the whole derivative into the explicit subset kernel  $K_{ab}^{(C)}$  in (2). The two views are complementary: the subset proof gives the cleanest global certificate, while the Jacobian proof explains the local mechanism.

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